

CSE 499 Project Plan

Project Name: Stomper

Team Members:

1. Gabriel Chagas Boff
2. Liezl Gonzaga Lizardo
3. Joshua Nathan Hyatt
4. Benjamin Walveranta

Requirements:

List your project requirements with accompanying user stories. All the “core” requirements must be completed. Optional enhancements are features you would like to add if time permits. For each requirement list the user stories you will be addressing.

Required

Use this table to document the features of your project. **You must have at least 4 requirements.** Modify the table as needed.

Requirement	*C/E	Description								
Live Packet Capture	C	Capture and parse live network packets from a selected network interface.								
		User Stories								
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Configurable Rule Engine	E	Allow custom detection rules using JSON/YAML configuration.						
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Email/Webhook Alerts	E	Send notifications when threats are detected.						
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*C=Core, E=Enhancement

Project Schedule:

Create a rough project schedule broken down by sprint. For each sprint list the milestones you will meet. This is a rough guideline for future planning, you will make a more detailed plan at the first of each sprint. You will not be held accountable to this schedule as it will likely change as the class progresses.

Project Schedule (Tentative)

Sprint 1

Goal: Project Setup and Research

Milestones:

- Install Rust and development tools
- Create GitHub repository
- Configure project structure
- Research packet capture libraries (pcap or pnet)
- Learn Rust basics
- Initial architecture design (Open to close)

Sprint 2

Goal: Packet Capture & Detection

Milestones:

- Capture live packets
- Parse Ethernet/IP/TCP/UDP headers
- Implement port scan detection
- Implement SYN flood detection

- Console alert generation
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Sprint 3

Goal: Storage & Dashboard

Milestones:

- SQLite integration
 - Store alerts
 - Build REST API (Axum)
 - Create dashboard
 - Display statistics and alert history
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Sprint 4

Goal: Testing & Finalization

Milestones:

- Bug fixes
 - Performance improvements
 - Documentation
 - User Guide
 - Presentation
 - Final demonstration
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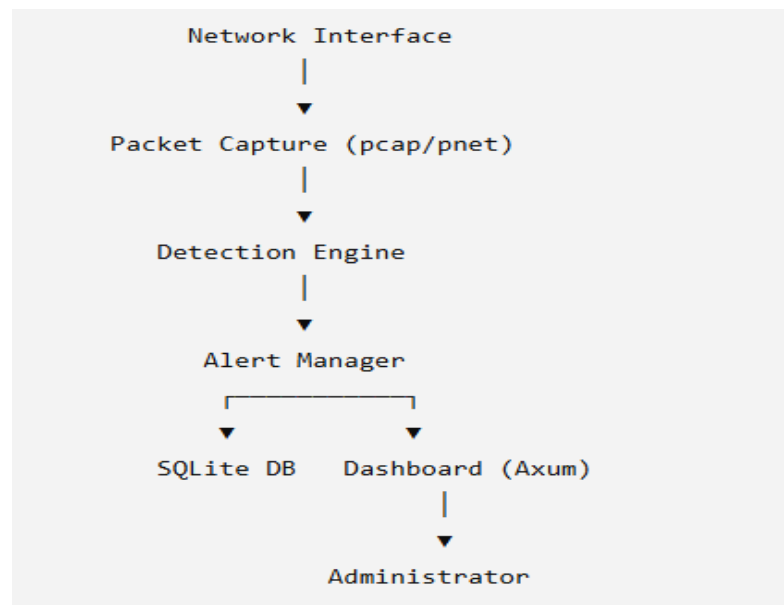
Tentative Team Responsibilities

Member	Responsibility
Gabriel	Packet capture and networking layer
Joshua	Detection engine and rule processing
Benjamin	Dashboard, APIs, and database integration
Liezl	Documentation (SRS, user stories, testing), GitHub coordination, UI support, SQLite integration, testing, and learning/contributing to Rust modules

Project Architecture

Stomper uses a client-server architecture with a modular processing pipeline to detect and monitor network intrusions in real time; the open to close is the principle that will be applied. The system begins by capturing live network packets from a selected network interface using Rust packet-capture libraries such as pcap or pnet. The captured packets are then passed to the Detection Engine, where they are analyzed for suspicious activities such as port scans, SYN flood attacks, and other malicious network behaviors.

If suspicious activity is detected, the Alert Manager generates an alert and records the event in a SQLite database for historical logging and analysis. The stored data is then displayed through a web-based dashboard built with Axum, allowing administrators to monitor network traffic, review security alerts, and view system statistics in real time. This modular architecture separates packet capture, threat detection, alert management, data storage, and the user interface into independent components, making the system easier to maintain, test, and enhance with additional detection features in the future.



Technology

Stomper will be developed using **Rust** as the primary programming language because of its high performance, memory safety, and built-in concurrency features, making it well-suited for network security applications. Live network traffic will be captured using **pcap** or **pnet**, Rust libraries that provide low-level access to network packets for real-time analysis.

The system will use the **Tokio** asynchronous runtime to support concurrent packet processing, allowing multiple network packets to be analyzed efficiently without blocking execution. Security alerts and historical event logs will be stored in a lightweight **SQLite** database using **rusqlite** or **sqlx**, providing persistent storage with minimal configuration.

A web-based dashboard will be developed using the **Axum** web framework to display traffic statistics, recent alerts, and system status. Configuration files and detection rules will be managed using **Serde**, enabling the application to read and write data in JSON or YAML format.

Development will be carried out in **Visual Studio Code** using the Rust Analyzer extension for code completion and debugging support. **Git** and **GitHub** will be used for version

control, collaboration, and source code management. The project will also use Rust's built-in testing framework to perform unit testing and verify the correctness of key system components throughout development.

Technology	Purpose
Rust	Core programming language for building the IDS
pcap / pnet	Capture and parse live network packets
Tokio	Asynchronous and concurrent packet processing
Axum	Web framework for the dashboard and API
SQLite	Store alerts and network event logs
rusqlite / sqlx	Connect Rust to the SQLite database
Serde	Read and write JSON/YAML configuration files
Visual Studio Code	Integrated development environment (IDE)
Git & GitHub	Version control and team collaboration
Rust Test Framework	Unit testing and code verification